When Does Lithium Replace Lead-Acid? A Temperature-Driven Techno-Economic, Carbon and Public-Health Assessment of Lead-Acid, LFP and NMC Batteries for Electric Easy-Bikes in Bangladesh

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[**TODO:** add co-authors; supervisor must approve]

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Abstract

Bangladesh's electric easy-bikes run overwhelmingly on lead-acid batteries that last about a year, waste grid electricity and feed an informal recycling chain that is a major source of lead exposure. Lithium iron phosphate (LFP) and nickel-manganese-cobalt (NMC) packs cost more up front, and their advantage under Bangladesh's hot climate has not been quantified. We build a transparent, temperature-driven MATLAB/Simulink model that couples NASA POWER 2 m air temperature (Khulna, Chuadanga, Rajshahi; 2024–2025) to capacity, internal resistance, depth of discharge, Arrhenius/Miner-rule ageing, replacement-aware 5- and 10-year discounted cost, and a carbon and lead-flow account. Inputs come from published sources wherever they exist (a 2026 field study of Khulna and Gopalganj easy-bikes, peer-reviewed ageing data, the June 2026 BERC tariff and an official grid emission factor); inputs without a source are run as explicit scenarios. Site-averaged results give a battery life of 12.4 months for lead-acid, 52.9 for NMC and 83.6 for LFP, and a 10-year net present cost of BDT 726,743 (lead-acid), 436,875 (NMC) and 341,625 (LFP) per bike. LFP cuts grid electricity by 15.2% and emits about 524 kg CO_2e less per bike-year than lead-acid ($\approx$0.73 Mt CO_2e yr^{-1} for 1.4 million bikes), while removing 82–170 kt of lead per year from the recycling stream. The three sites differ by only 0.32°C in mean battery temperature, so regional temperature does not change the ranking; battery temperature rise, daily distance and energy use matter more, and LFP's cost advantage disappears only when several unfavourable assumptions combine. The study is model-based and has not been validated against operator records.

Keywords: easy-bike; lead-acid; lithium iron phosphate; NASA POWER; battery degradation; total cost of ownership; carbon emissions; lead pollution; Bangladesh

1 Introduction

Battery-powered three-wheelers (easy-bikes, Mishuks, e-rickshaws) have become a mass transport mode in Bangladesh. A CLEAN/BWGED study estimates about 1.4 million easy-bikes consuming roughly 4,905 GWh of electricity per year [1], whereas other sources put the number of electric three-wheelers at 3–4 million [2]; we carry both figures as scenarios. Lead-acid batteries dominate [3] and are short-lived: a battery set costing more than BDT 72,000 is reported to last 8–11 months [4]. A 2026 field study that surveyed 150 drivers in Gopalganj and Khulna documents prolonged hub charging, frequent overloading and lead-acid battery degradation [5].

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This short life has three consequences. (i) *Economic*: replacement dominates operator cost. (ii) *Energy and carbon*: internal resistance and charger losses waste electricity, and each replacement carries manufacturing emissions. (iii) *Public health*: used lead-acid batteries (ULABs) are broken and smelted largely by the informal sector, a primary contributor to lead-pollution hotspots [6]. The national public-health bulletin of IEDCR summarises the burden: about 36 million Bangladeshi children (60%) are poisoned by lead, lead causes an estimated loss of about 20 million IQ points and 140,000 cardiovascular deaths, and the 2019 economic burden was US$28.6 billion, 6–9% of GDP [7]. Pure Earth reports about 125 kg of lead in one easy-bike's battery set and about 167,000 t of lead waste per year from the electric three-wheeler sector [2].

LFP and NMC batteries offer higher round-trip efficiency, much longer cycle life and no lead, but cost more and are imported. Bangladesh is also hot: monthly mean 2 m temperature in the study area ranges from 15.5 to 33.6°C (Figure 4), and a 42°C day has been reported in Chuadanga [8]. Heat accelerates ageing of every chemistry, but not equally, so whether lithium is preferable under local conditions, and by how much, is an empirical question.

Research gap. Existing work compares lead-acid and lithium-ion for stationary storage [9–11], studies cell ageing [12, 13], thermal safety [14], charging infrastructure [15], or easy-bike operation and energy use for lead-acid only [5, 16]. We are not aware of a study that links satellite-reanalysis temperature to chemistry-specific ageing, replacement-aware cost, carbon and lead-flow accounting for Bangladesh easy-bikes in one transparent model. [**TODO:** state the databases, search terms and dates of your final literature search]

Contributions.

1. A reproducible MATLAB/Simulink model (analytic and block-diagram versions cross-checked) of three chemistries driven by NASA POWER temperature.
2. A parameter matrix in which every value carries a published source or an explicit reason for being a scenario (Table 2).
3. A carbon and lead-flow account for the fleet, with fleet-size and emission-factor sensitivity.
4. A four-scenario policy comparison, a break-even analysis, and a worst-case test of when LFP stops being cheaper.

2 Literature Review and Parameter Matrix

2.1 Literature used

Table 1 lists the sources and what each contributes. Entries marked † were supplied in the author's data file and must be read in full before submission; the others were located and their key facts checked online (status in `references.bib`).

2.2 Parameter matrix

Table 2 lists every model input, the value used, its source and its status: *Sourced* (published or official figure), *Derived* (computed from sourced values), or *Scenario* (no public Bangladesh-specific source found; the value is varied in Section 4.4).

Table 1: Literature used, key finding relevant to this work, and the gap it leaves.

Source	Scope	Finding used here	Gap / how this paper adds
Chowdhury et al. [5]	Field survey (n=150, Gopalganj and Khulna) and one instrumented bike	5×12 V, 150 Ah lead-acid bank; 8–9 kWh per full charge; about 20 A at 60 V with five passengers; iron-core chargers about 80%; hub fee BDT 80–100 per charge	Lead-acid only; no chemistry or temperature model
CLEAN/BWGED [1]†	Fleet and grid load	≈1.4 M bikes; ≈4,905 GWh/yr	Fleet-level
Pure Earth [2]; IEDCR bulletin [7]	ULAB policy and lead burden	3–4 M e-three-wheelers; ≈125 kg Pb per bike; 36 M children poisoned	Not linked to battery life or chemistry
Yagci et al. [12]	LFP ageing	$E_a = 37.3 \pm 12.8$ kJ/mol	Stationary cells
Jalkanen et al. [13]	NMC/graphite cycle ageing	2600 cycles at room temperature vs 2000 at 45°C	One cell type
Tsujikawa et al. [17]; [18]	VRLA ageing	Life roughly halves per 8–10°C; damage summed over a variable temperature	Float service
Jasper et al. [11]	LCA, lead-acid vs LFP	Production+EOL about 40 (lead-acid) vs 50 (LFP) kg CO_2e/kWh; LFP better with many cycles	European data
Yudhistira et al. [10]; Paul et al. [19]	LCA grid storage; Li-battery LCA review	LFP lower impact per kWh delivered; NMC higher cell footprint	Not easy-bike duty
Keshan et al. [9]†	Off-grid stationary storage	Efficiency and cycle-life ranges for lead-acid and Li-ion	Stationary, not traction
Schöberl et al. [14]†	Thermal runaway	LFP onset > 270°C; NMC about 150–210°C	Safety only
Chowdhury et al. [20]; [21]	Remediation near ULAB sites	Soil lead −96%, child blood lead −35% (Savar); −22% (Tangail)	Health only

Table 2: Parameter matrix. LA = lead-acid. Model values are listed as LA / LFP / NMC where they differ.

Parameter	Model value	Source / reason	Status
Pack and performance			
Voltage	60 / 51.2 / 51.8 V	LA: five 12 V batteries [5]; LFP 16S, NMC 14S	Sourced / standard
Capacity	150 / 150 / 150 Ah (9.0 / 7.7 / 7.8 kWh)	LA 150 Ah bank [5]; Li packs sized to the same Ah so all can serve the duty	Derived
Round-trip efficiency	77.5 / 95 / 92.5%	Midpoints of ranges in the author's data file [9]	Sourced†
Cycle life at 80% DoD, 25°C	400 / 2,500 / 1,250	Midpoints of data-file ranges [9]; NMC lower bound consistent with [13]	Sourced†
Pack price (BDT)	90,909 / 162,000 / 170,455	Midpoint market prices in the data file per kWh, scaled to pack energy; LA reported >72,000 [4]	Sourced†, derived
Temperature effects			
Ambient temperature	15.5–33.6°C monthly mean	NASA POWER T2M, MERRA-2 [22, 23]; 42°C day [8]	Sourced
LA life vs heat	Halves per 8 K above 25°C	[18]; peer-reviewed 10 K [17] tested in Section 4.4	Sourced
Li ageing E_a	LFP 37.3; NMC 10.3 kJ/mol	LFP: 37.3 ± 12.8 [12]; NMC: from 2600 cycles (room T) vs 2000 (45°C) [13]	Sourced / derived
Capacity factor vs T	LA 0.80–1.12; Li 0.65–1.05 over 0–50°C	LA [24]; Li above 25°C [25]; cold side qualitative [26]; vendor-level, only winter months fall below 25°C	Vendor-level
Resistance sensitivity E_r	25 / 20 / 20 kJ/mol	No source found; small effect	Scenario
Cycle-life DoD exponent k	1.4 / 1.0 / 1.0	No source found; cycle life $\propto (0.8/DoD)^k$	Scenario
Battery above ambient	+5°C (0, 10 tested)	Not measured in [5]; *largest uncertainty*	Scenario
Duty cycle			
Energy use	79.7 Wh/km	Mean of 88 and 71.4 Wh/km from two studies [16]; 52–88 tested	Sourced
Average speed	15 km/h	1.2 kW (20 A×60 V, 5 passengers) [5] divided by 79.7 Wh/km	Derived
Daily distance	66 km	Calibrated so lead-acid grid energy is 8.5 kWh per charge, midpoint of 8–9 kWh [5]; 40–100 tested	Derived
Charger efficiency	80%	Iron-core chargers [5]	Sourced

Parameter	Model value	Source / reason	Status
Operating days	300 per year (250–350 tested)	No source found	Scenario
Economics			
Tariff	BDT 11.36/kWh (7.70, 16.24 tested)	BERC order of 3 June 2026 [27, 28]; wholesale revised again in August 2026 [29]; hub fee check: 80–100 BDT per 8–9 kWh ≈10.6 [5]	Sourced (secondary)
Discount rate	11.57% (±20% tested)	Weighted-average lending rate, July 2024 [30]; update to the latest Bangladesh Bank value	Sourced
Maintenance, scrap	0; scrap 0/15/30% of LA price	No Bangladesh price found; scenario runs	Scenario
Fleet, carbon and lead			
Fleet size	1.4 M [1]; 3.0 M	Sources disagree; both reported [2]	Sourced
Grid emission factor	0.62 tCO_2/MWh	Department of Environment, 2020–22 [31]; 0.55–0.70 tested	Sourced
Manufacture+EOL carbon	40 / 50 / 78.4 kg CO_2e/kWh	[11] (LA, LFP); NMC cell level [19]; LFP 50–110 tested	Sourced
Lead per pack	60 / 90 / 125 kg	125 kg [2]; 60 and 90 are scenarios	Sourced + scenario
Informal recycling share	70–80%	[2, 32, 33]	Sourced

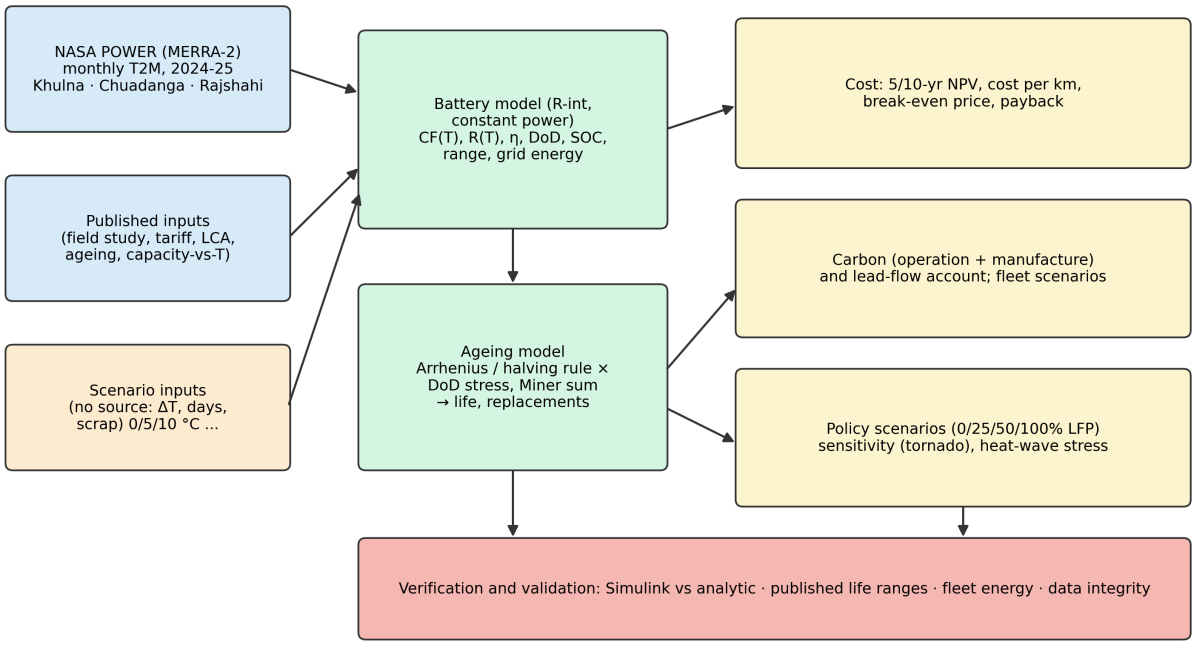


Figure 1: Methodology: temperature and parameter inputs drive the battery and ageing models; outputs feed the cost, carbon, lead and policy modules; verification and validation close the loop.

3 Methodology

3.1 Study design and sites

Three south-western sites are used: Khulna (22.84°N, 89.57°E), Chuadanga (23.64°N, 88.69°E) and Rajshahi (24.38°N, 88.60°E) (Figure 2). Chuadanga was selected because a 42°C day was reported there in May 2025 [8], giving a heat-stress case; note that in the NASA monthly means its annual mean (25.50°C in 2025) is not higher than Khulna's (25.59°C), so its selection rests on peak heat, not on the mean. Khulna is also one of the two districts of the field study that supplies the duty-cycle inputs [5]. Three chemistries are compared: lead-acid (LA), LFP and NMC. No operator survey was possible, so every duty-cycle input is taken from published sources or run as a scenario (Table 2).

3.2 Temperature data

Monthly mean 2 m air temperature (T2M, MERRA-2 [23]) for January 2024 to December 2025 was downloaded from NASA POWER [22]: 24 values per site, no missing (-999) entries. The native cell is $0.5° \times 0.625°$ (about 50–70 km), so the three points lie in neighbouring cells and are expected to be similar. Battery temperature is ambient plus $\Delta T_{op} = 5$°C, a scenario value tested at 0 and 10°C. Monthly means understate daily peaks (Section 5.6).

3.3 Battery model

Each pack is a resistive (R-int) battery with temperature-dependent capacity and resistance, discharged at constant average power $P_w = e_{km} v$. With pack voltage V and round-trip efficiency

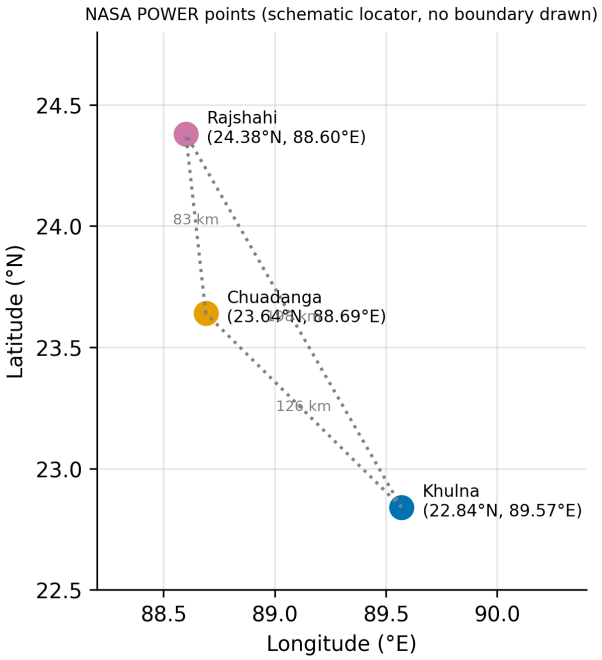


Figure 2: NASA POWER grid points used (schematic locator; distances are great-circle).

η_{RT}, the 25°C resistance is chosen so that discharge efficiency equals $\sqrt{\eta_{RT}}$ at 25°C:

$$R_{25} = \frac{(1-\sqrt{\eta_{RT}})\sqrt{\eta_{RT}}\,V^2}{P_w}, \quad R(T) = R_{25}\exp\left[\frac{E_r}{R_g}\left(\frac{1}{T_K}-\frac{1}{298.15}\right)\right], \quad I = \frac{V-\sqrt{V^2-4R(T)P_w}}{2R(T)}, \quad \eta_d = 1-\frac{}{} \tag{1}$$

The capacity factor $CF(T)$ is interpolated from a table, so the available energy is $E_{cap} = V\,Ah\,CF(T)/1000$ kWh. With daily delivered energy $E_{day} = d\,e_{km}/1000$ for distance d:

$$DoD = \frac{E_{day}}{\eta_d\,E_{cap}}, \qquad SOC_{end} = \max(0, 1-DoD), \tag{2}$$

$$E_{grid} = \frac{E_{day}}{\eta_d^2\,\eta_{chg}}, \qquad \text{Range} = \frac{DoD_{max}\,E_{cap}\,\eta_d}{e_{km}/1000}. \tag{3}$$

The same equations are implemented as a Simulink model built from core blocks (Figure 3): constant, lookup tables, product, a function block for the constant-power current, an integrator for SOC, and a sum. It is run at 15, 25, 35 and 45°C for all three chemistries and compared with the analytic code (`easybike_build_simulink.m`); the acceptance criterion is 0.5 percentage points in efficiency, 0.01 in end-of-day SOC and 2% in range. [**TODO:** run easybike_build_simulink in MATLAB and report the PASS table here]

3.4 Degradation and lifetime

Cycle life falls with depth of discharge and temperature, and damage is accumulated month by month with Miner's rule [34], as also done for variable-temperature VRLA service [17]:

$$N_{eff,m} = N_{80}\left(\frac{0.8}{\max(DoD_m, 0.05)}\right)^k f_T(T_m), \qquad D = n_{cyc}\sum_m \frac{w_m}{N_{eff,m}}, \qquad L = \frac{1}{D\,d_{op}}\ \text{(years)}, \tag{4}$$

where w_m is the day-weighted share of month m, $d_{op} = 300$ operating days per year and $n_{cyc} = 1$. For lithium, $f_T = \min\{1, \exp[\frac{E_a}{R_g}(\frac{1}{T_K}-\frac{1}{298.15})]\}$ with $E_a = 37.3$ kJ/mol (LFP, [12]) and 10.3 kJ/mol (NMC, derived from [13]); for lead-acid $f_T = \min\{1, 2^{-(T-25)/8}\}$ [18]. The cap at 1 gives no life credit below 25°C. Calendar ageing is not modelled, so life is cycle-limited (Section 5.6).

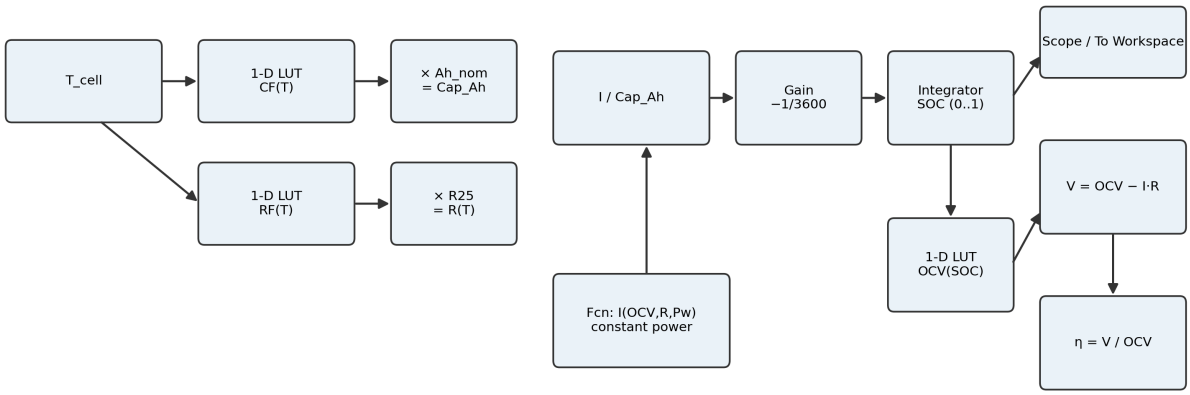


Figure 3: Block structure of the Simulink model `easybike_battery.slx` (schematic). [**TODO:** optional: replace by a screenshot of the model]

3.5 Total cost of ownership

The 5- and 10-year net present cost with discount rate r is

$$TCO_H = C_0 + \sum_{j=1}^{K} \frac{C_0(1-s)}{(1+r)^{jL}} - \frac{C_0\,\rho_H}{(1+r)^H} + \sum_{t=1}^{H} \frac{E_{grid}\,d_{op}\,\tau + m\,C_0}{(1+r)^t}, \tag{5}$$

with C_0 the pack price, $K = \lfloor H/L \rfloor$ replacements, s the scrap fraction, ρ_H the residual life fraction at horizon H, τ the tariff and m the maintenance fraction. Cost per km is TCO_{10} divided by the discounted distance. The cost is battery-related only (vehicle and driver costs are excluded).

3.6 Carbon, lead and policy

Annual emissions per bike are $E_{CO_2} = E_{grid}\,d_{op}\,EF_{grid} + \varepsilon\,Q_{nom}\,12/L_{mo}$ with $EF_{grid} = 0.62$ tCO$_2$/MWh [31], Q_{nom} the pack energy, L_{mo} the life in months and $\varepsilon = 40$, 50, 78.4 kg CO$_2$e/kWh (LA, LFP, NMC; [11, 19]). The NMC value is cell-level and the factors come from European studies; Li packs are imported, so LFP is tested up to 110 kg CO$_2$e/kWh. A second estimate scales the CLEAN fleet electricity by the modelled LFP-to-LA grid-energy ratio. Informal-smelting emissions have no verified factor and are excluded, which biases the lead-acid footprint downward. Lead cycled per year is $M_{Pb} = N\,m_{Pb}\,12/L_{mo}$ with $m_{Pb} = 60$, 90, 125 kg per pack: a mass flow, not an emission or a dose. Policy scenarios move a share $\phi = 0$, 25, 50, 100% of a 1.4 M fleet from LA to LFP; outputs scale linearly.

3.7 Validation and sensitivity

Five checks are made: V1 data integrity; V2 energy use against published values (a 110 km-per-charge benchmark of five 120 Ah batteries [35] and 71.4–88 Wh/km [16]); V3 modelled life against reported ranges (8–12 months for lead-acid [4]; 60–84 and 36–48 months for LFP and NMC from the data file); V4 grid energy against the 8–9 kWh per charge of the field study [5] (used for calibration, so not independent) and against the CLEAN fleet figure (independent); V5 Simulink against the analytic model. One-at-a-time sensitivity (11 inputs, Figure 7) and scenario runs for battery temperature, distance, scrap value and tariff address inputs without a published source.

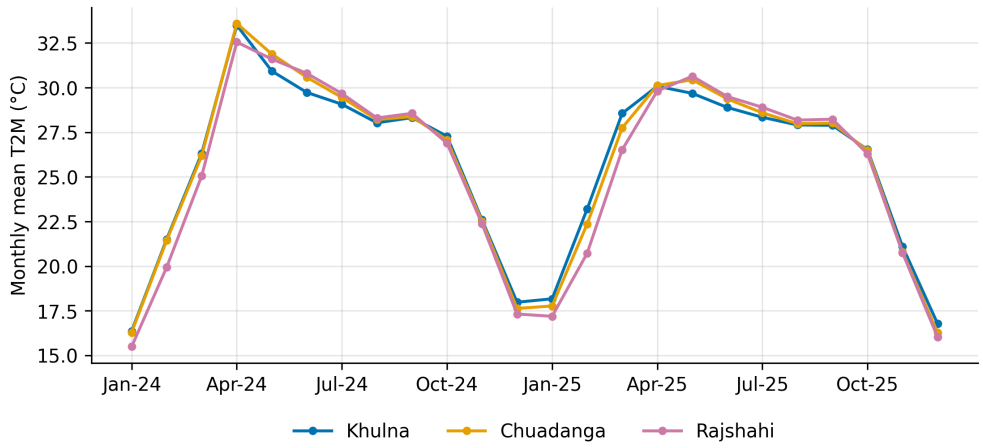


Figure 4: NASA POWER monthly mean 2 m temperature at the three sites, Jan 2024–Dec 2025.

Table 3: Simulated 24-month results (NASA POWER monthly T_{2m}, Jan 2024–Dec 2025; battery temperature = ambient + 5 °C). Source: `results_summary.csv` from `easybike_main.m`. Life is in months of operation at 300 days yr^{-1}; TCO is net present value at an 11.57% discount rate and a tariff of BDT 11.36/kWh.

Site	Battery	$\bar{T}_{cell}$ (°C)	η (%)	DoD (%)	Range (km)	E_{grid} (kWh/d)	Life (mo)	Repl. (10 yr)	TCO_{10} (10^3 BDT)	Cost (BDT/km)
Khulna	Lead-Acid	30.8	90.2	63.5	83.3	8.10	12.4	9	725	6.37
	LFP	30.8	97.8	69.4	76.1	6.87	83.7	1	342	3.00
	NMC	30.8	96.7	69.5	76.0	7.03	52.9	2	437	3.84
Chuadanga	Lead-Acid	30.8	90.2	63.5	83.3	8.10	12.2	9	731	6.42
	LFP	30.8	97.8	69.4	76.1	6.87	83.2	1	342	3.01
	NMC	30.8	96.7	69.5	76.0	7.03	52.8	2	437	3.84
Rajshahi	Lead-Acid	30.5	90.0	63.7	83.0	8.13	12.4	9	724	6.36
	LFP	30.5	97.8	69.5	76.0	6.88	84.0	1	341	2.99
	NMC	30.5	96.7	69.6	75.9	7.04	52.9	2	437	3.84

4 Results

4.1 Temperature input

Monthly mean T2M follows the same seasonal cycle at all three sites, from about 15.5°C in January to 33.6°C in April 2024 (Figure 4). Mean battery temperature is 30.8, 30.8 and 30.5°C at Khulna, Chuadanga and Rajshahi, a spread of 0.32°C (Table 3).

4.2 Performance, life and cost

Table 3 and Figure 5 give the main comparison. Discharge efficiency is about 90.1% (LA), 97.8% (LFP) and 96.7% (NMC). Modelled life is 12.4 months for LA, 52.9 for NMC and 83.6 for LFP (site mean), i.e. about 9 LA packs but 1 LFP pack replacement in ten years. The 10-year net present cost per bike is BDT 726,743 (LA), 436,875 (NMC) and 341,625 (LFP): LFP saves BDT 385,118 per bike, and its cost per km (3.00 BDT) is less than half that of LA (6.38 BDT). The LA bank has the longest range (83 vs 76 km) because it stores more energy (9.0 vs 7.7 kWh), not because it is more efficient. All packs run close to the 80% DoD limit in some months (maximum 70/72/74%), which is realistic for easy-bike duty.

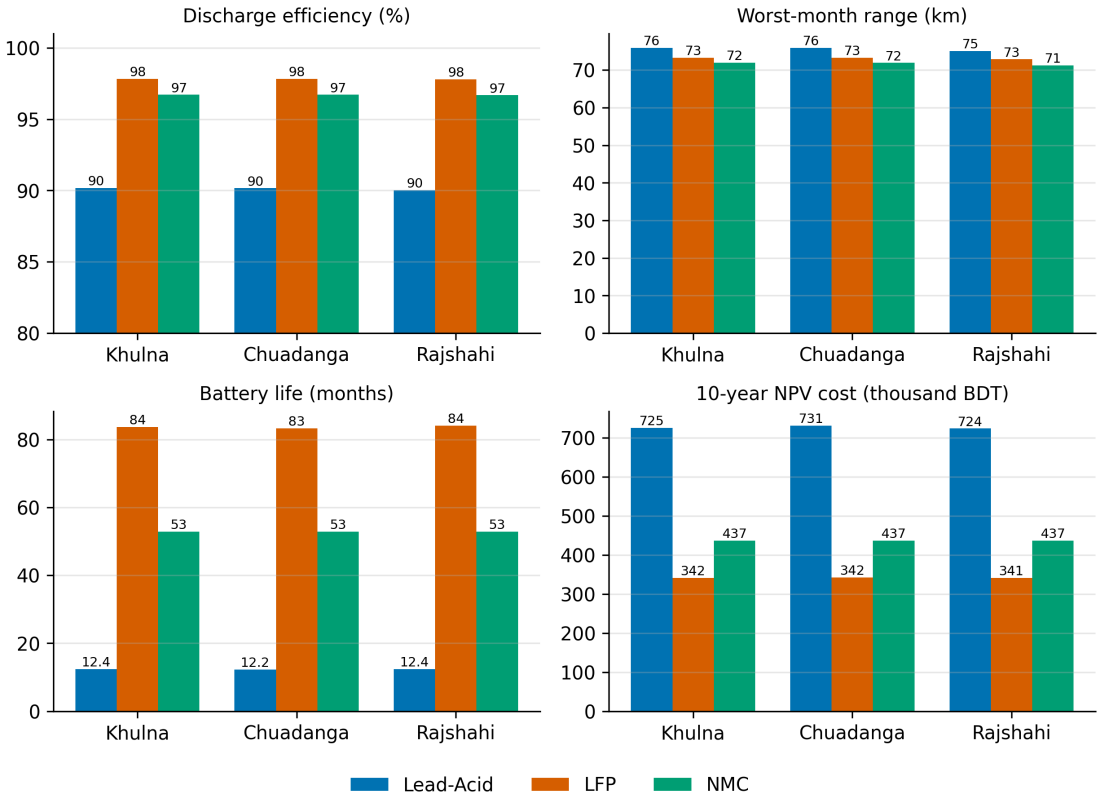


Figure 5: Site comparison of discharge efficiency, worst-month range, battery life and 10-year NPV cost.

4.3 Heat-wave stress

At 42°C ambient (Chuadanga, 10 May 2025 [8]) with the 5°C offset, LA ages 3.2 times, LFP 2.0 times and NMC 1.2 times faster than in the 2024–25 monthly-mean climate. The absolute life gap between lead-acid and lithium therefore widens in heat.

4.4 Scenarios, sensitivity and break-even

Table 4 shows that the battery temperature rise and daily distance change life and cost strongly. A battery 10°C above ambient cuts LA life to 8.7 months, while 0°C raises it to 16.3. At 100 km per day none of the three packs can serve the duty within 100% DoD, and at 40 km per day all last far longer. The tornado chart (Figure 7) ranks daily distance, battery temperature rise and energy use above price, tariff or ageing parameters. LFP stays cheaper than LA up to an LFP pack price of BDT 465,522 (model price 162,000), BDT 328,779 at 40 km per day and BDT 563,043 at +10°C. The up-front premium of BDT 71,091 is repaid in about 1.0 year of running savings (undiscounted).

When does LFP stop being cheaper? We combined the settings most unfavourable to LFP: 40 km per day, no battery heating, LA scrap value 30% of price, LA halving every 10 K, and an LFP price 20% above the model value. LFP is still cheaper by BDT 21,782. If the LFP price is 50% higher in that same case, LA becomes cheaper by BDT 22,555. LFP's cost advantage is therefore robust to any single assumption but not to a combination of several, and the LA scrap value is the most important missing datum (Table 4 and the tornado: a 30% scrap value cuts the gap by BDT 242,601).

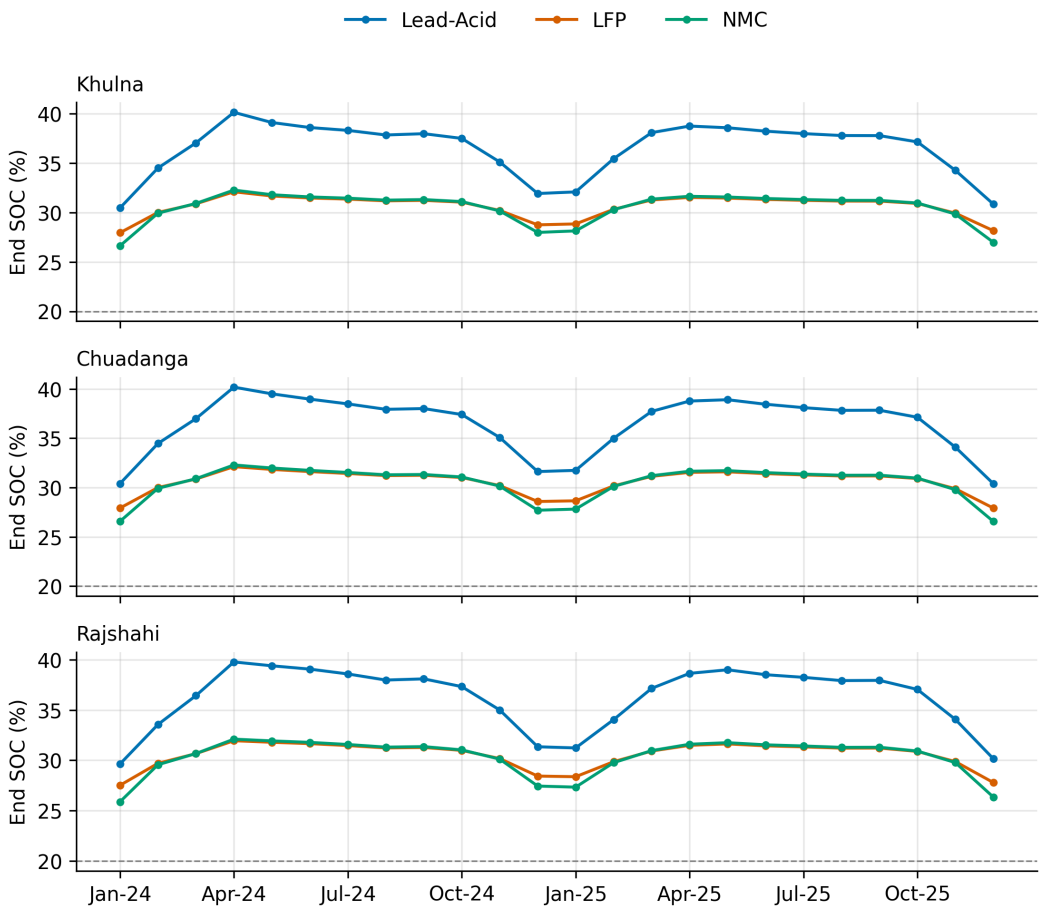


Figure 6: Month-end state of charge after one day's discharge; dashed line = 20% limit.

4.5 Validation

V1 passed (24 months per site, no gaps). V2 passed: 79.7 Wh/km lies inside the 52–88 Wh/km spanned by the published values. V3 (Figure 8): LFP life (83.6 mo) lies at the upper edge of the 60–84 reported; LA life (12.4 mo) is 3.0% above the 8–12 months field range; NMC life (52.9 mo) is 10% above 36–48. V4: modelled lead-acid grid energy (8.1 kWh per bike-day) is 16% below the CLEAN-implied 9.6 kWh, a gap that only field metering can resolve (the CLEAN figure may include other vehicle types and charging losses). V5 is the Simulink cross-check described in the Methodology. The model is therefore plausible against the published ranges but is *not* validated against operator records.

4.6 Carbon footprint

A lead-acid easy-bike causes about 1,858 kg CO_2e per year, an LFP bike 1,334 and an NMC bike 1,447 (Table 5 and Figure 9). Switching one bike to LFP saves 524 kg CO_2e per year, 230 from lower electricity use and 295 from fewer manufactured packs (0.97 LA packs against 0.14 LFP packs per year). For 1.4 M bikes this is 0.73 Mt CO_2e per year, or 1.57 Mt for 3 M bikes (Table 6). The independent CLEAN-based estimate, electricity only, gives 747 GWh and 0.46 Mt CO_2 per year. The per-bike saving stays between 432 and 554 kg per year for LFP manufacturing between 50 and 110 kg CO_2e/kWh and grid factors of 0.55–0.70 (Figure 10).

Table 4: Scenario runs for inputs without a published source. Upper block: battery temperature rise above ambient (site mean). Lower block: daily distance at Chuadanga (max DoD >100% means the pack cannot serve the duty).

	Life (months)			10-yr TCO (10^3 BDT)		
	LA	LFP	NMC	LA	LFP	NMC
ΔT = +0 °C	16.3	98.9	54.0	604	321	433
ΔT = +5 °C	12.4	83.6	52.9	727	342	437
ΔT = +10 °C	8.7	68.1	50.7	950	370	446
40 km/day (max DoD 42/44/45%)	24.7	137.3	87.1	397	237	295
60 km/day (max DoD 63/66/67%)	14.0	91.5	58.1	649	317	402
66 km/day (max DoD 70/72/73%)	12.2	83.2	52.8	731	342	437
80 km/day (max DoD 84/87/89%)	9.4	68.7	43.6	932	398	515
100 km/day (max DoD 105/109/111%)	6.8	54.9	34.9	1240	480	627

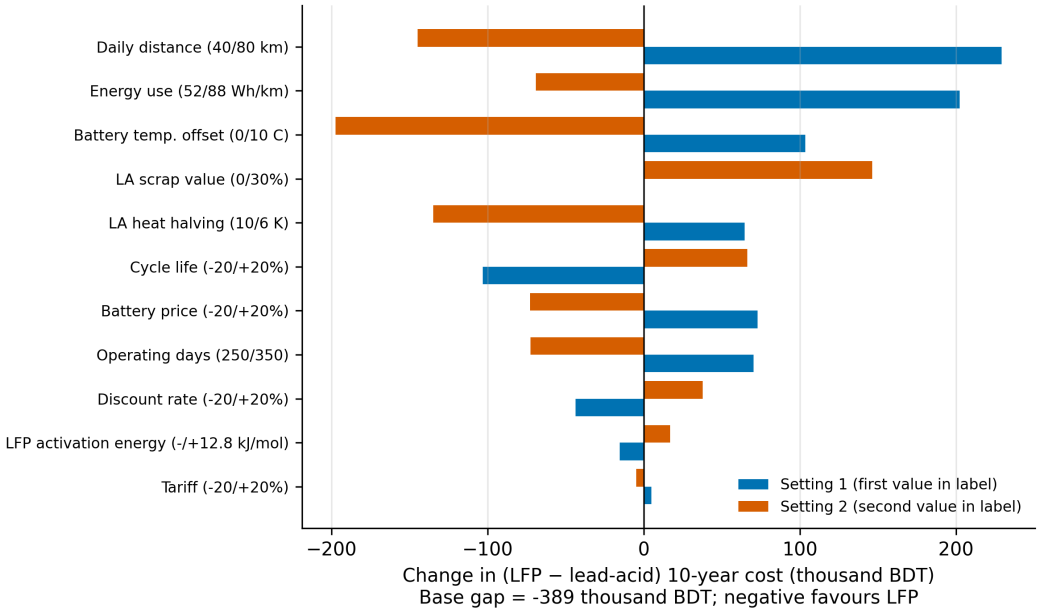


Figure 7: One-at-a-time sensitivity of the LFP-minus-LA 10-year cost gap at Chuadanga (base gap −388,897 BDT). Negative values favour LFP.

4.7 Lead flow and policy scenarios

The lead-acid fleet cycles 82–170 kt of lead per year at 1.4 M bikes (Table 7); the upper value is close to the 167 kt reported by Pure Earth for the whole electric three-wheeler sector [2]. Converted bikes need replacement so rarely that their lead flow in the model falls to zero. A 50% conversion saves 0.37 Mt CO_2e per year and BDT 270 billion for operators over ten years, and full conversion saves 0.73 Mt per year and BDT 539 billion (Table 8 and Figure 11).

5 Discussion

5.1 Why LFP wins in this model

Three mechanisms act together. First, LFP's higher round-trip efficiency cuts charging energy by 15.2%. Second, ageing at 30–34°C is far slower per cycle for lithium, so one LFP pack outlasts about seven lead-acid packs. Third, fewer replacements mean fewer manufactured packs and less lead entering the recycling chain. The lead-acid life (12.4 months) is close to the 8–12 months

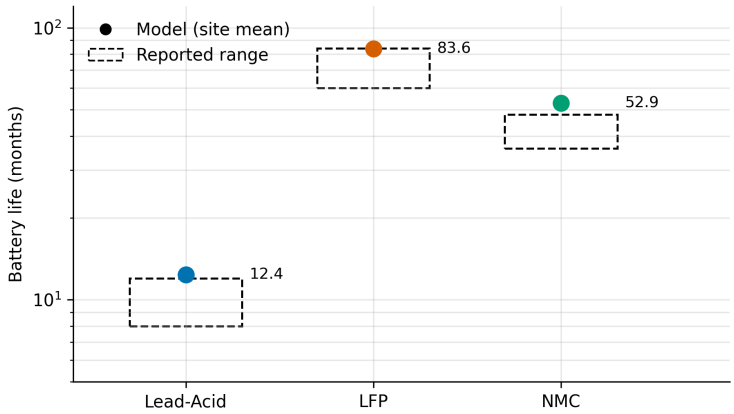


Figure 8: Modelled battery life (site mean) against reported ranges (dashed boxes; logarithmic axis).

Table 5: Carbon budget per bike per year (site-averaged; grid factor 0.62 tCO_2/MWh; manufacture+end-of-life factors per kWh of pack capacity: 40 (lead-acid), 50 (LFP), 78.4 (NMC) kg CO_2e).

Battery	Packs/yr (–)	Operation (kg CO_2e)	Manufacture+EOL (kg CO_2e)	Total (kg CO_2e)
Lead-acid	0.97	1508	350	1858
LFP	0.14	1279	55	1334
NMC	0.23	1308	138	1447

reported in the field [4], so the baseline is realistic; the LFP life of 83.6 months rests on laboratory-derived ageing laws and a cycle-life range, and has not been observed in Bangladeshi service, which is why operator data remain the most valuable next step. The ranking agrees with LCA findings that LFP is preferable where many daily cycles occur [10, 11].

5.2 Temperature: what matters and what does not

Regional temperature does *not* separate Khulna, Chuadanga and Rajshahi: mean battery temperatures differ by 0.32°C and every ranking is identical. Chemistry choice is therefore robust to site within the south-west. The temperature effect operates through the seasonal cycle, heat waves, and above all the battery's temperature rise above ambient (Table 4), the largest single uncertainty. Measuring pack temperature in service is a higher priority than adding weather stations.

5.3 Carbon: a smaller prize than health, but robust

A full switch saves about 0.73 Mt CO_2e per year at 1.4 M bikes, and the saving holds across manufacturing footprints of 50–110 kg CO_2e/kWh and grid factors of 0.55–0.70. As the grid decarbonises the operational saving falls while the embodied saving remains. The carbon case alone is not the strongest argument for switching; cost and lead are stronger.

5.4 Lead, workers and public health

Used lead-acid batteries are collected, broken open, and their lead plates smelted, often by hand and close to homes and farmland. Pure Earth describes informal recycling that contaminates and poisons workers and nearby residents [2]; ESDO identifies it as a primary contributor to lead-pollution hotspots [6]. The IEDCR bulletin reports that about 36 million children (60%) are poisoned by lead, and that market screening in Dhaka, Rajshahi, Khulna and Barishal found 24% of 197 consumer samples above reference lead levels; home assessments in 47 Khulna households

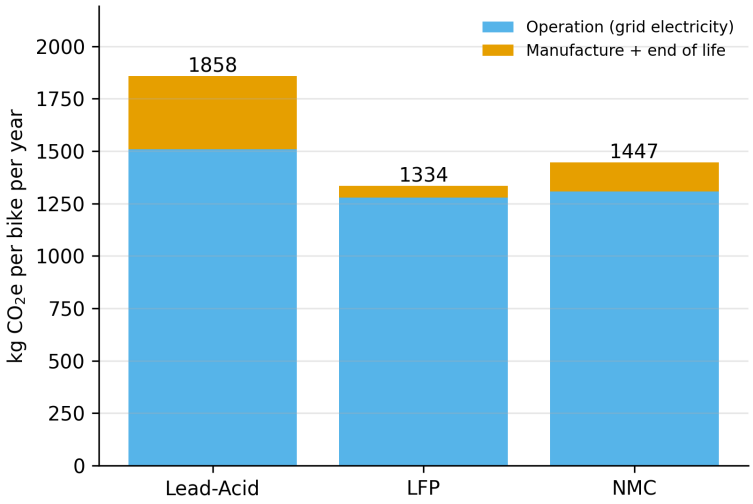


Figure 9: Annual carbon footprint per bike, split into operation and manufacture+end-of-life.

Table 6: Fleet-level annual CO_2e under two fleet-size scenarios; savings are lead-acid minus the alternative if the *entire* fleet switched.

	1.4 M bikes [1] (Mt CO_2e/yr)	3.0 M bikes (lower bound of 3–4 M [2]) (Mt CO_2e/yr)
Lead-acid fleet (baseline)	2.60	5.57
Saving, all LFP (operation)	0.32	0.69
Saving, all LFP (manufacture+EOL)	0.41	0.88
Saving, all LFP (total)	**0.73**	**1.57**
Saving, all NMC (total)	0.58	1.23

found high lead in metal and ceramic cookware and toys [7]. Children are most vulnerable: the World Health Organization notes that no safe level of lead exposure is known [36]. Remediation helps but only partly: soil lead fell 96% at an abandoned Savar site while child blood lead fell 35% [20]; the IEDCR bulletin reports a 20% fall at Mirzapur, where hotspot soil dropped from about 100,000 ppm to 60–100 ppm [7]; and a Tangail study reports 22% [21]. The fact that blood lead falls much less than soil lead suggests other sources persist, so removing the source of new contamination is as important as cleaning old sites. Between 70% and 80% of ULAB recycling is reported to be informal [2, 32, 33], and e-rickshaws and easy-bikes generate about 77% of ULABs [37].

Our model supplies the *quantity* side: by extending battery life from about 12 to about 84 months, an LFP conversion removes 82–170 kt of lead per year from the recycling stream at 1.4 M bikes. It does not estimate how much of that lead would reach people; that needs emission and exposure data that do not yet exist, and other sources (paint, spices, cookware) also contribute.

5.5 Policy implications

(i) *Up-front finance*: the BDT 71,091 premium is repaid in about 1.0 year, so the barrier is capital access, not total cost; pay-as-you-go and battery-swap schemes address this [38]. (ii) *Taxes and duties*: current taxes on lithium batteries are described as unfavourable [38], while the break-even price of about BDT 465,522 leaves large room for relief. (iii) *ULAB regulation*: the 2021 Hazardous Waste (e-waste) Management Rules reportedly cover household but not e-rickshaw lead batteries [38]. (iv) *Lithium end-of-life*: a collection and recycling route must exist before volumes grow.

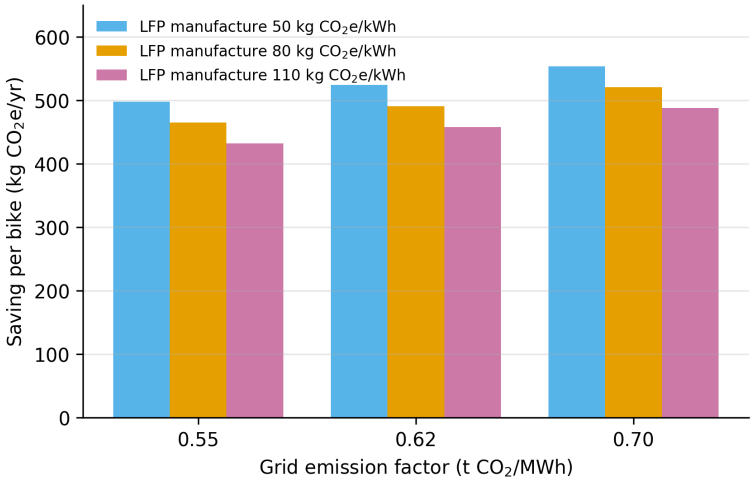


Figure 10: Per-bike saving from LFP versus lead-acid for three LFP manufacturing footprints and three grid emission factors.

Table 7: Lead throughput of the lead-acid fleet (kt Pb per year through replacement and recycling). 125 kg is the figure published by Pure Earth [2]; 60 and 90 kg are scenarios (battery mass × lead share).

Lead per pack	1.4 M bikes	3.0 M bikes
60 kg	82	175
90 kg	122	262
125 kg	170	364

5.6 Limitations

- **No operator records.** Battery temperature rise, operating days and scrap value are scenarios; daily distance is calibrated to a published energy per charge. The model is plausible against published ranges but not validated against operator data.
- **Cycle-limited life.** Calendar ageing is not modelled; the LFP life of 83.6 months (about 7 years) may be optimistic in hot, rough service, though the 10-year cost includes only one replacement.
- **Vendor-level capacity curves** for the temperature dependence of capacity; the effect is small between 20 and 45°C.
- **Monthly temperature.** Hourly data would capture daily peaks and charging-time heat.
- **Single-site field study.** Duty inputs come from one instrumented bike and a survey in Gopalganj and Khulna [5].
- **Fleet size.** 1.4 M and 3–4 M are both reported; results scale linearly.
- **Weight, fire safety and imports** are not modelled.
- **Lead flow is not exposure.**
- **Tariff.** The BDT 11.36/kWh rate is from the June 2026 order as reported by the press; wholesale prices were revised again in August 2026 [29]. Tariffs of 7.70–16.24 change the cost gap by less than 3% (Table 4).

6 Conclusion

Using published data and a transparent, reproducible model, LFP batteries outperform lead-acid for Bangladesh easy-bikes on efficiency (97.8 vs 90.1%), life (about 83.6 vs 12.4 months), 10-year cost (BDT 341,625 vs 726,743) and carbon (about 524 kg CO_2e per bike-year). NMC lies between the two. The ranking is the same at Khulna, Chuadanga and Rajshahi, so site temperature does

Table 8: Policy scenarios for a 1.4 M-bike fleet: share moved from lead-acid to LFP. Lead flow assumes 90 kg Pb per pack (60–125 kg, Table 7); savings are 10-year NPV to operators.

Scenario	LFP share (%)	CO_2e saved (Mt/yr)	Pb cycled (kt Pb/yr)	Operator saving (10^9 BDT)
S0 Business as usual	0	0.00	122	0
S1 Early adoption	25	0.18	92	135
S2 Targeted policy	50	0.37	61	270
S3 Full transition	100	0.73	0	539

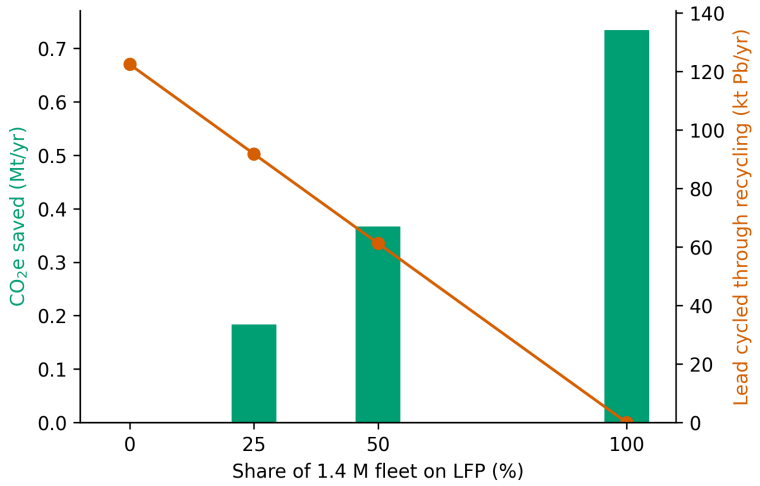


Figure 11: Policy scenarios: CO_2e saved and lead cycled through recycling versus the share of the fleet on LFP.

not change it, while battery heating and heat waves widen the lead-acid disadvantage. A full switch of a 1.4 M fleet would save about 0.73 Mt CO_2e per year and remove 82–170 kt of lead per year from informal recycling. The premium is repaid in about a year, pointing policy toward financing, taxes and recycling rules. LFP's cost advantage is robust to any single input but can disappear when several unfavourable assumptions combine; operator records on distance, pack temperature and scrap value are the priority for confirming it.

7 Reproducibility

Run in MATLAB R2024a: (1) `easybike_main` (tables and figures); (2) `easybike_co2` (carbon, lead and policy); (3) `easybike_build_simulink` (Simulink verification; needs Simulink); (4) `easybike_check` (compares your `results_summary.csv` with the values in this paper). Inputs live in `easybike_params.m`, each tagged with its source key. NASA POWER files: Monthly, T2M, 2024-01-01 to 2025-12-31, local solar time, points 22.84°N 89.57°E, 23.64°N 88.69°E and 24.38°N 88.60°E. Tables and figures in this paper were generated by a Python port that reproduced a previous MATLAB `results_summary.csv` to 15 significant digits; the MATLAB code in the final form must be run once to confirm Tables 4–8.

Declarations

Data availability. NASA POWER files, MATLAB/Simulink code (`easybike_*.m`, `easybike_battery.slx`) and all result tables are provided as supplementary material. [**TODO:** deposit on Zenodo/GitHub and add DOI]

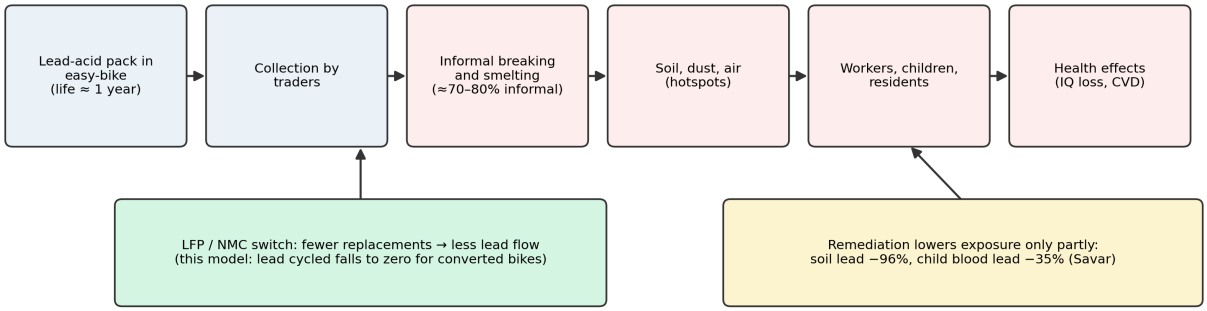


Figure 12: Pathway from lead-acid easy-bike batteries to exposure (schematic) and the point where an LFP switch acts; remediation figures from [20].

Funding / Conflicts of interest. [**TODO:** complete]
Acknowledgements. [**TODO:** complete]

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A Data and parameter files

`matlab/expected_results.csv` holds the nine site-battery rows behind Table 3. `easybike_check.m` reports whether your MATLAB output matches them.